

BIRD SPECIES OBSERVATION ANALYSIS

A Data Analytics Internship Project Report

Task- 4

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Internship Project Report

1. Project Overview

This project focuses on analyzing bird species observation data to understand biodiversity patterns, environmental influences, and conservation importance.

The main objectives of this project were:

- To clean and preprocess raw data
- To perform exploratory data analysis (EDA)
- To extract insights using SQL queries
- To build an interactive dashboard for visualization

This project follows a complete data analytics pipeline from raw data to insights.

2. Dataset Description

The dataset contains bird observation records with the following attributes:

- **Location_Type** – Forest / Grassland
- **Common_Name & Scientific_Name** – Bird species
- **Temperature & Humidity** – Environmental factors
- **Distance** – Observation range
- **Sky** – Weather condition
- **Observer** – Person who recorded observation
- **Flyover_Observed** – Movement indicator
- **PIF_Watchlist_Status** – Conservation status
- **Admin_Unit** – Location

The dataset enables multi-dimensional ecological analysis.

3. Data Cleaning & Preprocessing

What is happening?

The raw dataset contained missing values, inconsistent formats, and multiple sheets.

Why is it happening?

Data was collected from different sources and stored in separate sheets.

What was done?

- Missing values were handled
- Data types were corrected
- Multiple sheets were merged into a single dataset
- Column names were standardized

What does it mean?

The dataset became clean, consistent, and ready for analysis.

4. Exploratory Data Analysis (EDA)

4.1 Temporal Analysis

What is happening?

The dataset contains observations from only one year.

Why is it happening?

Data collection was limited to a single year.

What does it mean?

Year-wise trend is not meaningful; other time-based analysis is preferred.

4.2 Spatial (Habitat) Analysis

What is happening?

Forest areas show higher bird observations compared to grasslands.

Why is it happening?

Forests provide better ecological conditions such as food, shelter, and nesting.

What does it mean?

Habitat type significantly affects biodiversity.

4.3 Species Analysis

What is happening?

A few species dominate the observations.

Why is it happening?

Some species are more common or easier to detect.

What does it mean?

Species distribution is uneven.

4.4 Environmental Analysis

What is happening?

Bird activity varies with temperature and humidity.

Why is it happening?

Birds are more active under favorable environmental conditions.

What does it mean?

Environmental factors influence bird behavior and visibility.

4.5 Behavior Analysis**What is happening?**

Flyover observations indicate bird movement.

Why is it happening?

Birds pass through locations without staying.

What does it mean?

Data includes both presence and movement patterns.

4.6 Observer Analysis**What is happening?**

Some observers recorded more observations than others.

Why is it happening?

Different levels of observation effort.

What does it mean?

Potential observer bias exists in the dataset.

4.7 Conservation Analysis**What is happening?**

Watchlist species are present in the dataset.

Why is it happening?

Some species require monitoring and protection.

What does it mean?

Certain areas are important for conservation.

5. SQL-Based Analysis

SQL was used to perform structured analysis on the dataset.

Queries Performed:

- Habitat-wise observation count
- Top bird species
- Average temperature
- Species diversity by habitat

- Observer activity
- Distance distribution
- Weather condition analysis
- Habitat-wise temperature
- Watchlist species count
- Top observation locations

What is happening?

Data is analyzed using structured SQL queries.

Why is it happening?

SQL enables efficient and fast data aggregation.

What does it mean?

SQL results validate and support EDA findings.

6. Dashboard Development

An interactive dashboard was built using Streamlit.

Features:

- Theme toggle (Light/Dark mode)
- Filters (Habitat, Year, Species)
- Interactive charts (Bar, Pie, Scatter, Line)
- SQL insights integration
- Data download functionality

What is happening?

Users can interactively explore the dataset.

Why is it happening?

Visualization simplifies complex data.

What does it mean?

The dashboard improves usability and decision-making.

7. Key Insights

- Forest habitats show higher biodiversity than grasslands
- Environmental conditions influence bird activity
- Some species dominate observations

- Certain locations act as biodiversity hotspots
- Presence of watchlist species highlights conservation importance

8. Conclusion

This project successfully demonstrates the use of data analytics techniques to analyze bird observation data.

Key outcomes:

- Clean and structured dataset
- Meaningful insights using EDA
- Efficient analysis using SQL
- Interactive dashboard for visualization

The project provides a complete end-to-end data analysis workflow and highlights the importance of biodiversity and conservation.